For Performance Measurement

# ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

## General Certificate of Education Advanced Level

**PHYSICS** **6032/1**

PAPER 1 Multiple Choice 1 hour

**SPECIMEN PAPER**

Additional materials:
Multiple Choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)
Electronic calculator

**INSTRUCTIONS TO CANDIDATES**

Answer **all** questions. For each question, there are **four** possible answers, **A**, **B**, **C** and **D**. Choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet.

**Read very carefully the instructions on the answer sheet.**

**INFORMATION FOR CANDIDATES**

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

**This specimen paper consists of 17 printed pages.**

# DATA

| | |
|---|---|
| speed of light in free space | $c = 3.00 \times 10^{8} ms^{-1}$ |
| permeability of free space | $\mu_o = 4\pi \times 10^{-7}\ Hm^{-1}$ |
| permittivity of free space | $\varepsilon_0 = 8.85 \times 10^{-12}\ Fm^{-1}$ ( $1/4\pi\varepsilon_0 = 8.99 \times 10^{9}\ mF^{-1}$) |
| elementary charge | $e = 1.60 \times 10^{-19}$ C |
| the Planck constant | $h = 6.63 \times 10^{-34}$ Js |
| unified atomic mass unit | $1\ u = 1.66 \times 10^{-27}$ kg |
| rest mass of electron | $m_e = 9.11 \times 10^{-31}$ kg |
| rest mass of proton | $m_p = 1.67 \times 10^{-27}$ kg |
| molar gas constant | $R = 8.31\ JK^{-1}mol^{-1}$ |
| the Avogadro constant | $N_A = 6.02 \times 10^{23} mol^{-1}$ |
| the Boltzmann constant | $k = 1.38 \times 10^{-23}\ JK^{-1}$ |
| gravitational constant | $G = 6.67 \times 10^{-11} Nm^2kg^{-2}$ |
| acceleration of free fall | $g = 9.81\ ms^{-2}$ |

## FORMULAE

| | |
|---|---|
| uniformly accelerated motion | $s = ut + \frac{1}{2}at^2$ |
| | $v^2 = u^2 + 2as$ |
| work done on/by a gas | $W = p\,\Delta V$ |
| gravitational potential | $\emptyset = -\text{Gm/r}$ |
| hydrostatic pressure | $p = \rho\text{gh}$ |
| pressure of an ideal gas | $p = \frac{1}{3}\frac{Nm}{V} < c^2 >$ |
| simple harmonic motion | $a = -\omega^2 x$ |
| velocity of particle in s.h.m. | $v = v_o \cos \omega\text{t}$ |
| | $v = \pm\omega\sqrt{(x_o^2 - x^2)}$ |
| Attenuation of xrays | $I = I_o e^{-\mu x}$ |
| electric potential | $V = \frac{Q}{4\pi\varepsilon_0 r}$ |
| capacitors in series | $1/C = 1/C_1 + 1/C_2 + \ldots$ |
| capacitors in parallel | $C = C_1 + C_2 + \ldots$ |
| energy of charged capacitor | $W = \frac{1}{2}QV$ |
| electric current | $I = Anvq$ |
| resistors in series | $R = R_1 + R_2 + \ldots$ |
| resistors in parallel | $1/R = 1/R_1 + 1/R_2 + \ldots$ |
| Hall voltage | $V_H = \frac{BI}{\text{ntq}}$ |
| alternating current/voltage | $x = x_o \sin\omega t$ |
| radioactive decay | $x = x_o \exp(-\lambda t)$ |
| decay constant | $\lambda = \frac{0.693}{t_{\frac{1}{2}}}$ |

1 Which one is a vector?

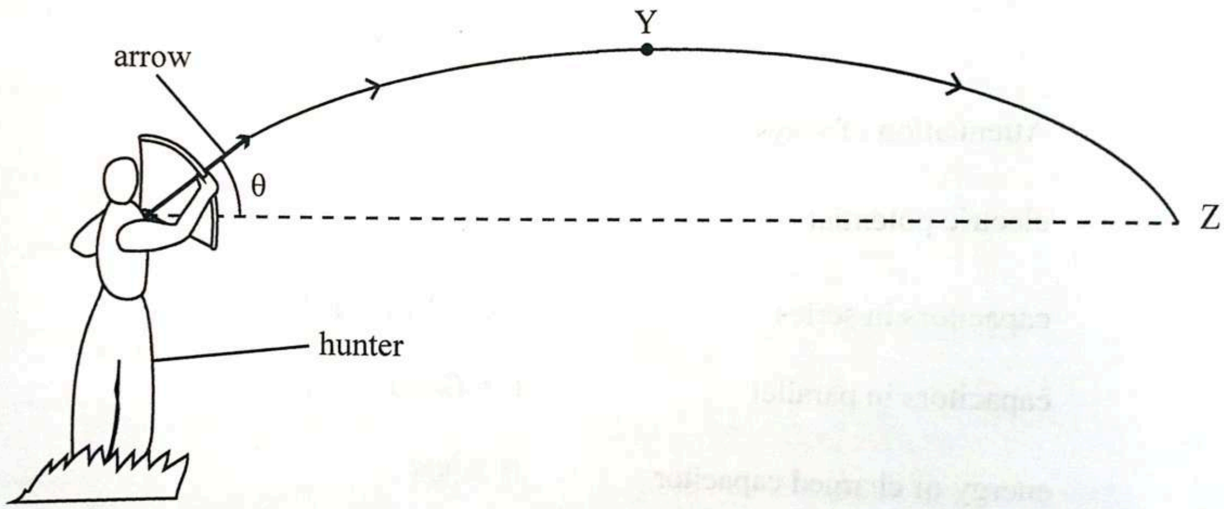
- **A** electric field strength
- **B** electric potential
- **C** magnetic flux
- **D** Young's Modulus

2 The percentage errors in the measurements of radius, height and mass of a cylinder are 1%, 2% and 6% respectively.

What is the percentage error in the density of the cylinder?

- **A** 12%
- **B** 10%
- **C** 9%
- **D** 8%

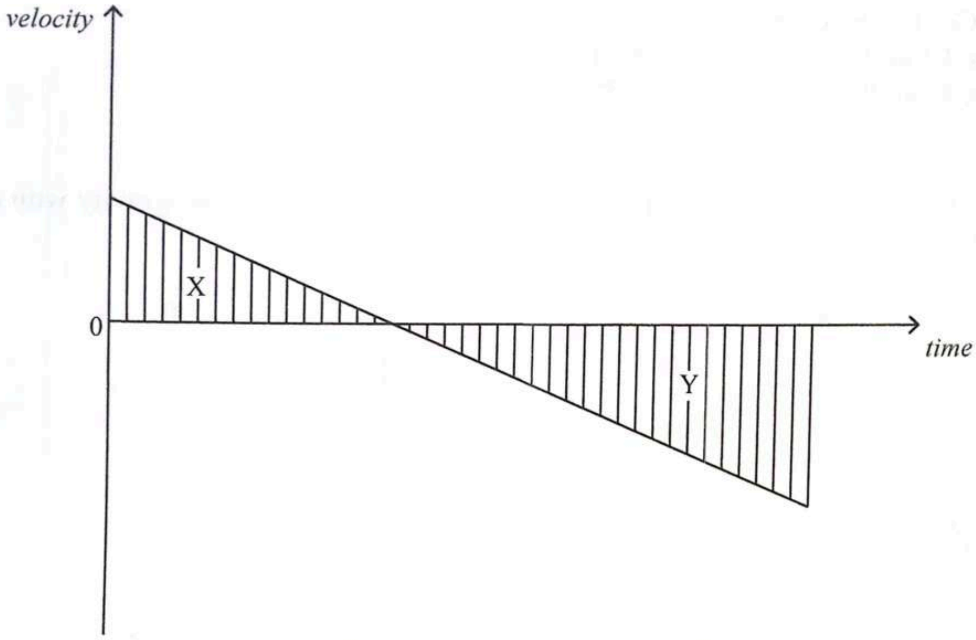
3 A hunter throws an arrow so that it follows a parabolic path as shown in the diagram.




Which statement is correct?

- **A** The horizontal velocity is zero at Y.
- **B** The horizontal velocity and vertical velocity of the arrow are constant.
- **C** The initial vertical component of velocity is zero.
- **D** The horizontal velocity and vertical acceleration are constant.

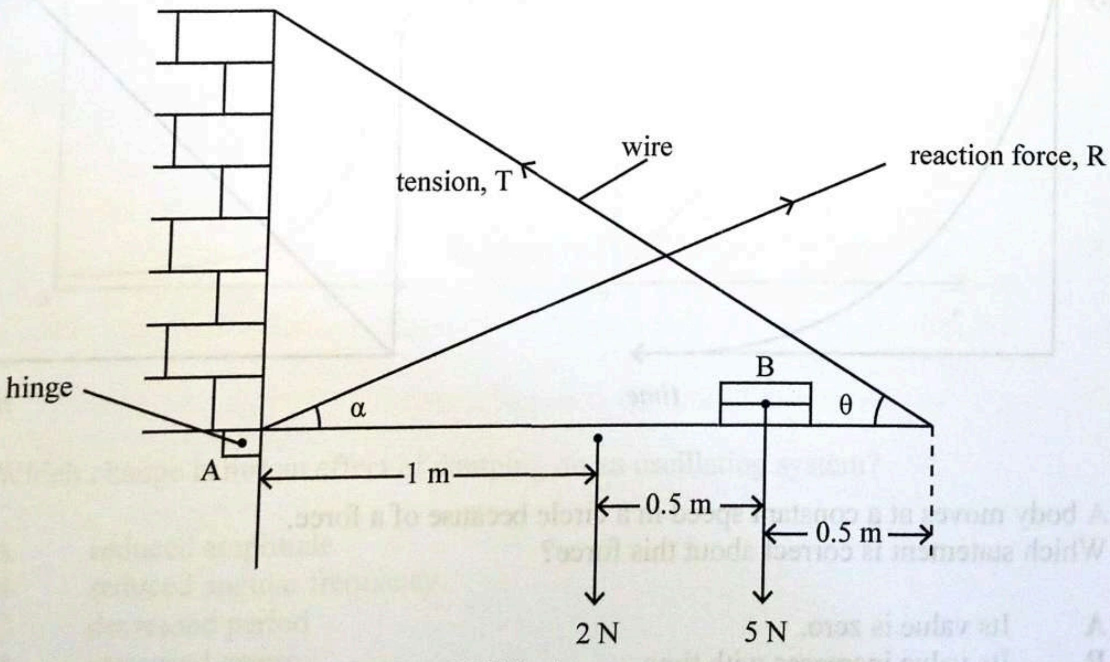
**4** An object is thrown vertically upwards from a cliff.
The *velocity-time* graph shows its motion until it lands below the point of release.




Which area represents the distance fallen below the cliff top?

**A** X + Y
**B** X
**C** Y
**D** Y - X

**5** The diagram shows a uniform plank 2 m long hinged at A and supports a pile of books placed at B. The wire holds the system in equilibrium.

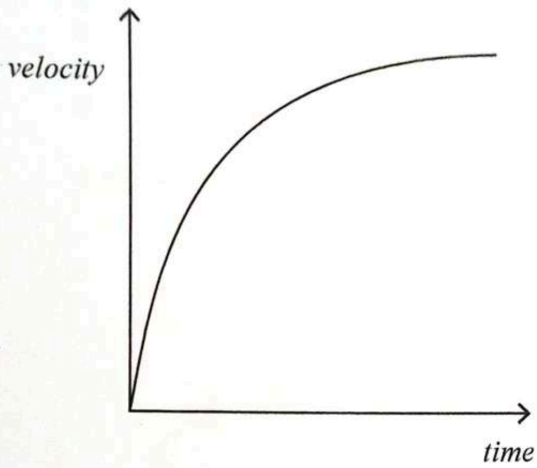


Which equation is correct for the system?

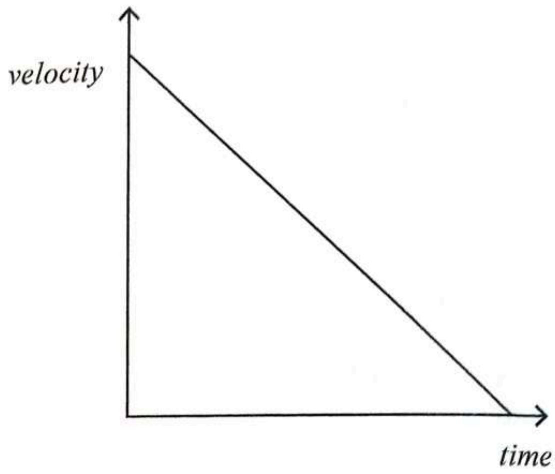
**A** $T \sin\theta + R \cos\alpha = 7$
**B** $T \cos\theta + R \sin\alpha = 7$
**C** $2 \times T \sin\theta = (2 \times 1) + (1.5 \times 5)$
**D** $2 \times T \sin\theta = (2 \times 1) + (0.5 \times 5)$

**6** Which *velocity-time* graph best represents a body falling from rest under gravity with air resistance?

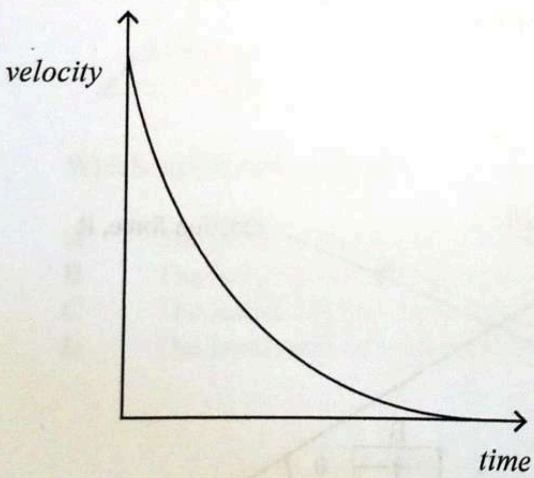
**A**



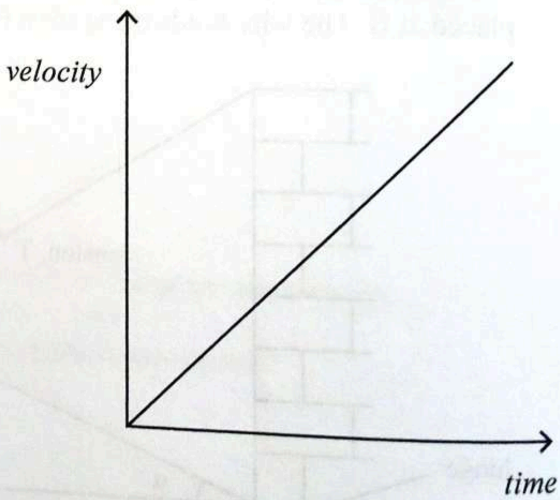

**B**




**C**




**D**

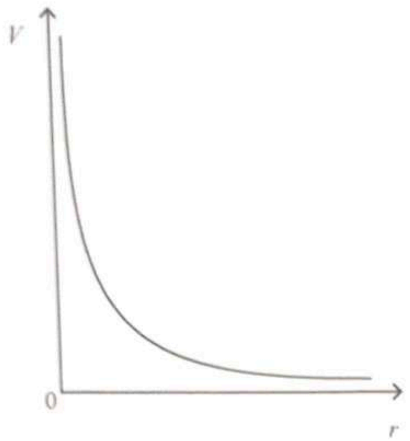



**7** A body moves at a constant speed in a circle because of a force.
Which statement is correct about this force?

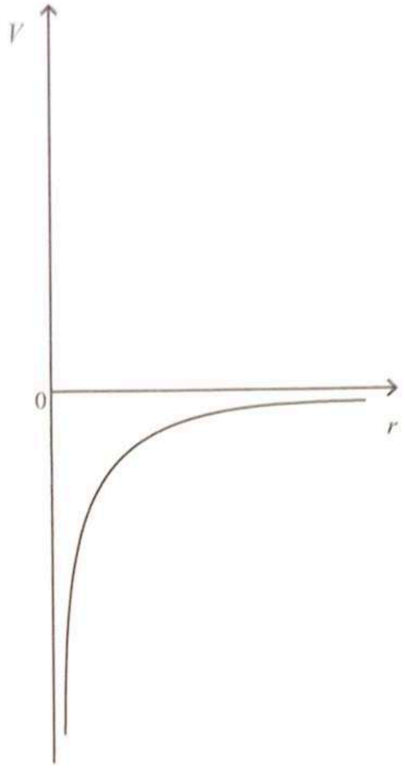
**A** Its value is zero.
**B** Its value increases with time.
**C** It is proportional to the speed of the body.
**D** It is proportional to the square of the velocity of the body.

8 Which graph shows the variation of gravitational potential, $V$, with distance, $r$, above the Earth's surface?

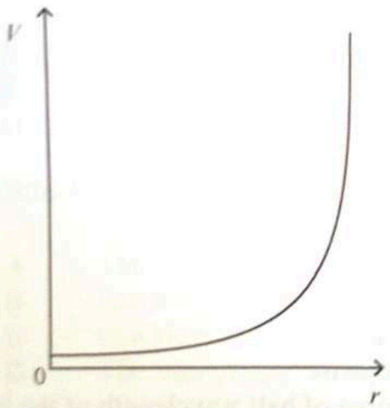
A



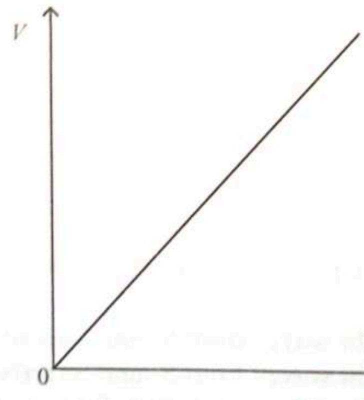

B




C




D




9 Which change is **not** an effect of damping on an oscillating system?

**A** reduced amplitude
**B** reduced angular frequency
**C** decreased period
**D** increased energy

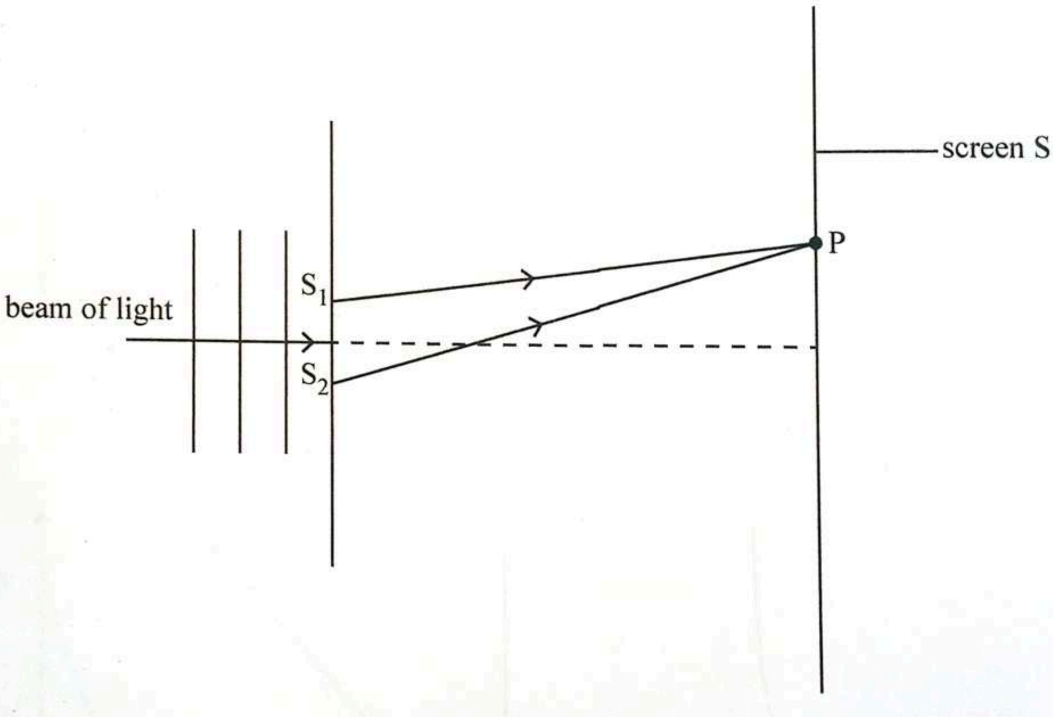
10 Which statement is correct about energy in a stationary wave?

**A** It is lost as heat.
**B** It is stored between nodes.
**C** It is transferred along the wave.
**D** It oscillates between nodes and antinodes.

11 What is a voxel in CT scanning?

**A** an X- ray beam
**B** a 2– dimensional pixel
**C** smallest 2 D unit of an image
**D** a small cube in a three-dimensional image

12 The diagram shows a beam of light passing through slits $S_1$ and $S_2$ to a screen, S.



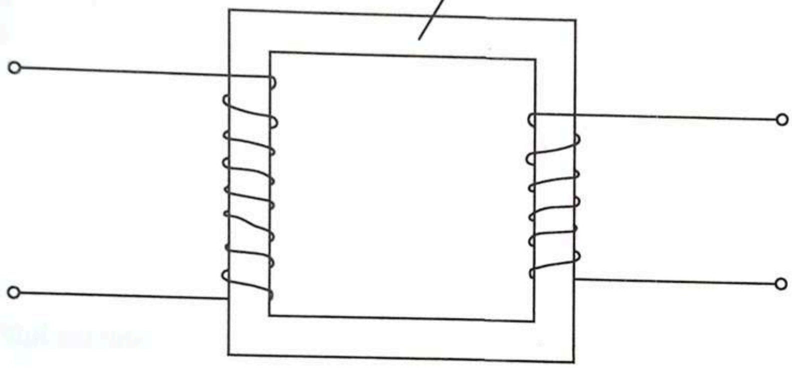

A bright fringe is observed at P because

**A** the light waves from $S_1$ and $S_2$ arrive at P in phase.
**B** the light waves from $S_1$ and $S_2$ arrive at P in antiphase.
**C** the path difference $S_2P$- $S_1P$ is equal to odd number of half wavelength of the beam of light.
**D** the two waves from $S_1$ and $S_2$ are not coherent.

13 Maxima produced by a diffraction grating are brighter than fringes produced in a double slit experiment because

**A** in a double slit experiment, fewer fringes are produced.
**B** in a double slit experiment, more light is scattered.
**C** a grating has more slits.
**D** a grating is made of glass.

14 The diagram shows a simple iron cored transformer.




The purpose of the continuous loop is

**A** to reduce power loss in the transmission cables.
**B** to increase magnetic flux linkage.
**C** to reduce hysteresis.
**D** to reduce resistance in the coil.

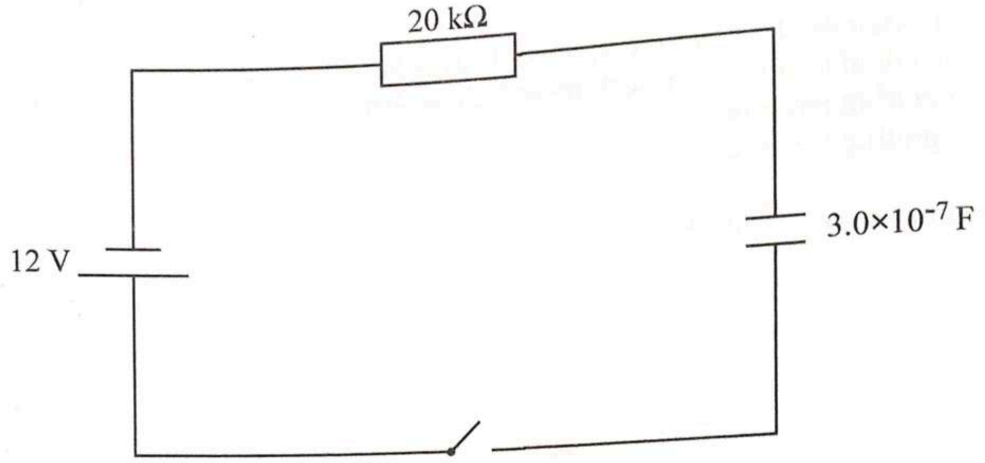
15 Which statement describes Kirchhoff's second law?

**A** The sum of e.m.f's is equal to the sum of p.d.'s.
**B** In a loop the sum of internal resistances is equal to the sum of p.d's.
**C** In a closed loop the sum of e.m.f.'s is equal to the sum of p.d.'s.
**D** The sum of p.d.'s is equal to the product of resistances in a closed loop.

16 Two parallel plates are 12 $\mu$m apart and have a potential difference of 2.5 kV. What is the electric field strength across the plates?

**A** $2.08 \times 10^{8}$ $Vm^{-1}$
**B** $2.08 \times 10^{5}$ $Vm^{-1}$
**C** $2.08 \times 10^{11}$ $Vm^{-1}$
**D** $2.08 \times 10^{2}$ $Vm^{-1}$

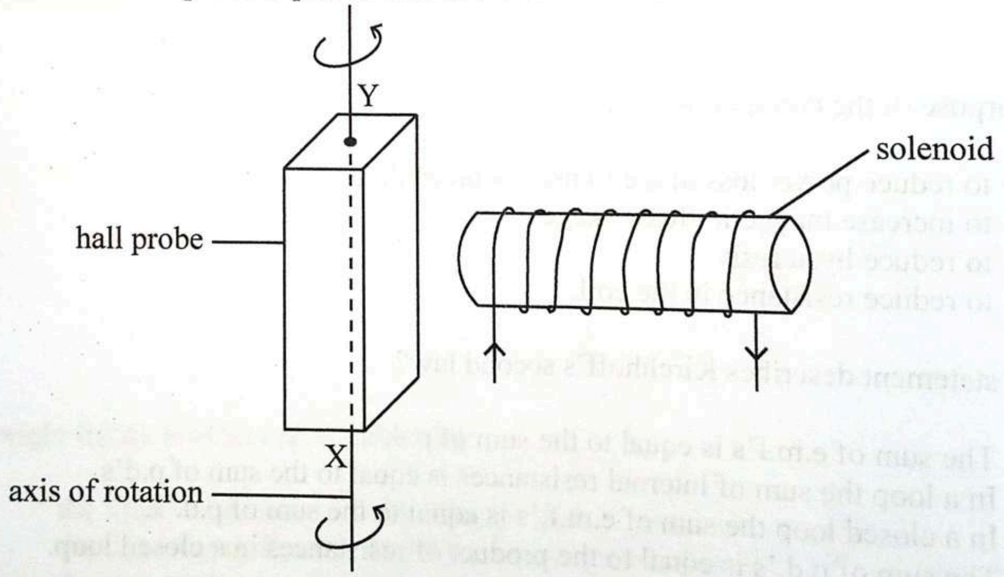
**17** The diagram shows an RC circuit.




What is the maximum charge stored by the capacitor?

**A** $2.5 \times 10^{-8}$ C
**B** $3.6 \times 10^{-6}$ C
**C** $6.6 \times 10^{-6}$ C
**D** $6.5 \times 10^{-3}$ C

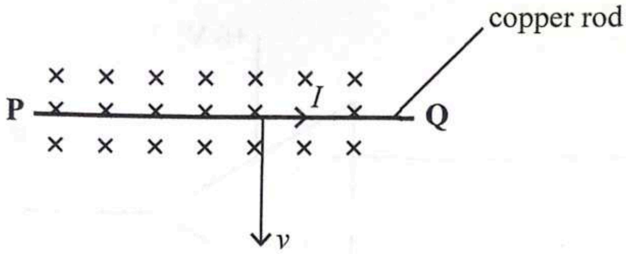
**18** A calibrated hall probe is placed near one end of a solenoid and rotated about the line XY.




Which statement explains why the hall voltage varies?

**A** Hall p.d is independent of the angle between the plane of the probe and magnetic field line.
**B** Hall p.d reduces to zero when the field is perpendicular to the plane and maximum when the field is parallel to the plane.
**C** Hall voltage depends on the maximum angle between the plane of probe and the magnetic field.
**D** Hall voltage attains its maximum value when the plane of probe is perpendicular to field lines and zero when parallel to field lines.

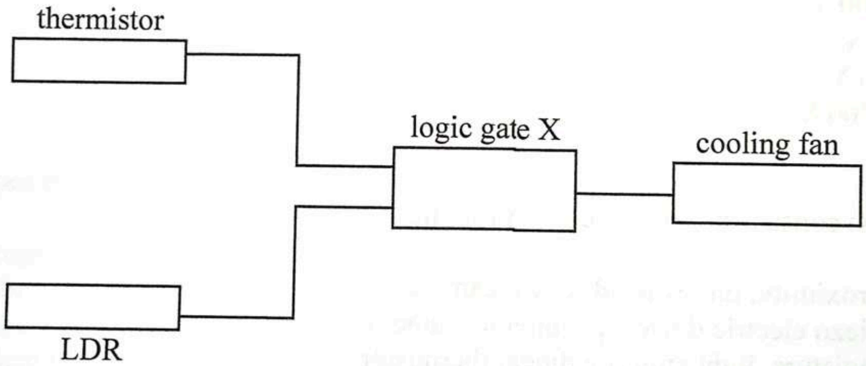
19 The diagram shows a copper rod moving with speed, $v$, in a magnetic field.




A current, $I$, flows from **P** to **Q** so that

**A** it produces a force that opposes the movement of the rod.
**B** it produces a force that increases the speed of the rod.
**C** it produces a force to accelerate charge carriers in the rod.
**D** it produces a force that opposes the movement of charge carriers in the rod.

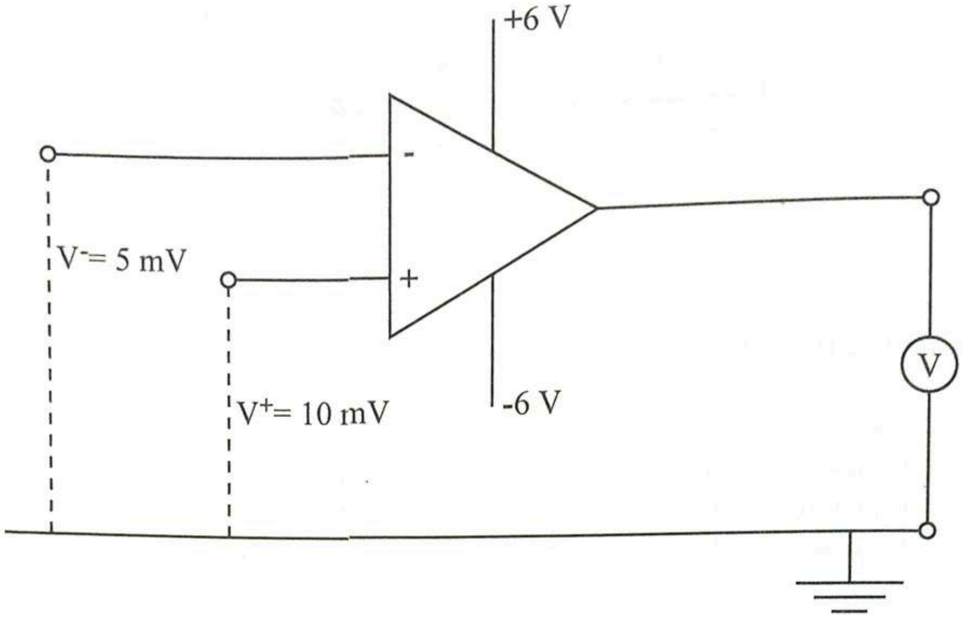
20 The block diagram shows a cooling fan controlled by a logic gate X.




The thermistor has a logic state 1 when it is hot and the LDR has a logic state I in daylight.
What is the logic gate X that will switch on the cooling fan in daylight when it is hot?

**A** NOR
**B** AND
**C** EX-OR
**D** NAND

21 The diagram shows an operational amplifier circuit.




What is the voltmeter reading?

**A** 500 V
**B** 6 V
**C** -6 V
**D** -500 V

22 Which list consists of sensors used in robotics?

**A** proximity, ultrasonic devices, camera
**B** piezo electric device, pneumatic, camera
**C** moisture, light emitting diode, thermistor
**D** electric motor, thermocouple, ultrasonic device

23 Which advantage makes pneumatic actuators suitable for simple robotic tasks in local industries?

**A** high precision and accuracy
**B** low cost and simplicity
**C** ability to handle heavy loads
**D** fast response

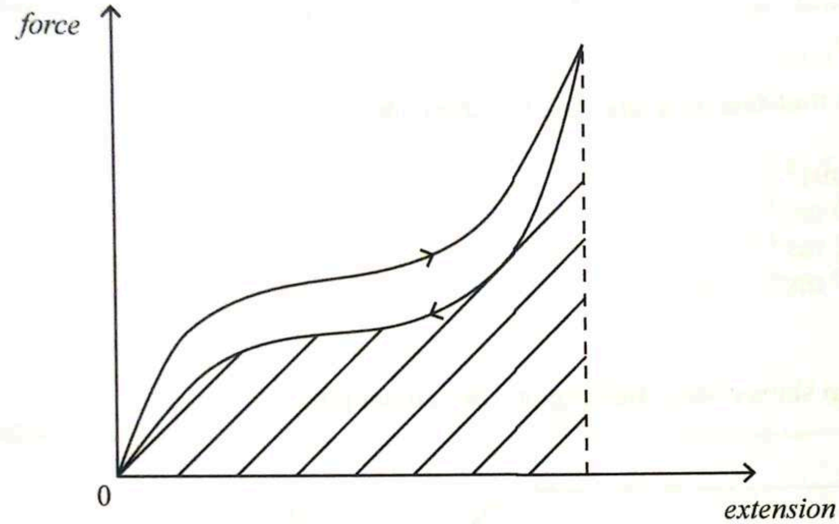
24 In Aurdino programming, what does the value 9 600 set?

**A** It sets the digital pin 9 to high and 6 to low.
**B** It determines the baud rate for serial communication.
**C** It sets up Arduino board clock's frequency at 9600Hz.
**D** It sets the digital pin 9 to low and digital pin 6 to zero.

25 Which phase transition occurs over a range of temperatures rather than at a fixed temperature?

A melting
B freezing
C evaporation
D boiling

26 The graph shows a *force-extension* curve for loading and unloading of rubber.




What does the shaded area represent?

A regained energy after unloading
B the work done when loading
C the Young modulus of rubber
D heat lost during a complete cycle

27 Two fixed points of a thermodynamic scale are

A 0°C and -100 °C.
B 0°C and 273. 0 °C.
C 0 K and 373.15 K.
D 0 K and -273.15 K.

28 Which statement is correct when a balloon containing an ideal gas is compressed at constant temperature?

A Work is done on a gas and heat energy goes into the gas.
B Work is done on a gas and energy goes out of the gas.
C Work is done by the gas and heat energy goes into the gas.
D Work is done by the gas and heat energy goes out of the gas.



[Turn over

29 Total change in thermal energy of a system equals heat added minus the

- **A** internal energy of the system.
- **B** work is done by the system.
- **C** kinetic energy of the system.
- **D** heat capacity of the system.

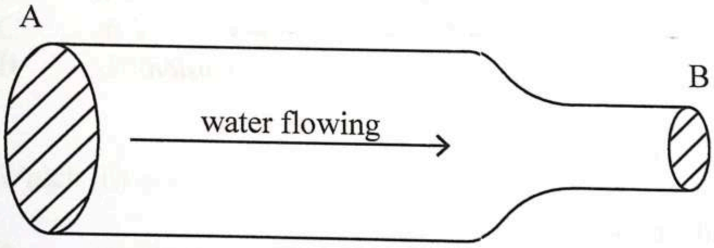
30 The table shows the speed of seven gas molecules recorded at a particular time.

| | | | | |
|---|---|---|---|---|
| number of particles | 1 | 3 | 2 | 1 |
| speed ($ms^{-1}$) | 5.0 | 6.5 | 7.0 | 7.5 |

What is the root-mean-square speed of these molecules?

- **A** 6.6 $ms^{-1}$
- **B** 24.6 $ms^{-1}$
- **C** 43.2 $ms^{-1}$
- **D** 43.7 $ms^{-1}$

31 The diagram shows water flowing in a horizontal pipe.




The pressure due to water between points A and B is

- **A** greater at A.
- **B** greater at B.
- **C** same at A and B.
- **D** lower at A.

32 Emission of radiation is done by

- **A** stars such as the sun only.
- **B** any objects whose temperature is above 0K.
- **C** objects with higher specific heat capacity only.
- **D** objects whose temperature is greater than their surroundings.

33 Which one is a deduction from Millikan's oil drop experiment?

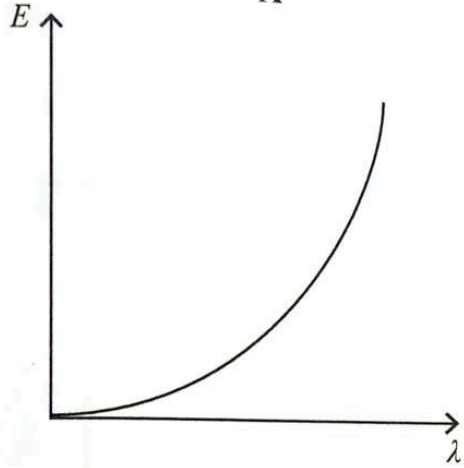
- **A** quantisation of energy
- **B** quantisation of charge
- **C** quantisation of momentum
- **D** quantisation of mass

**34** Which statement is correct about monochromatic light falling on a metal surface causing photoelectric emission?

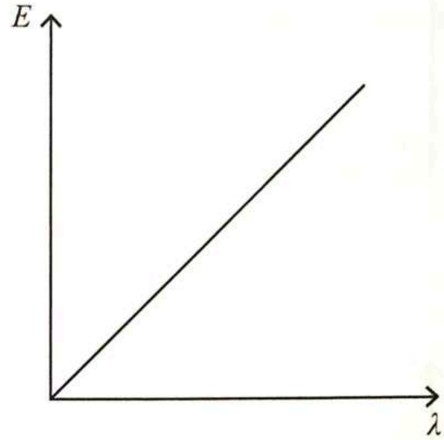
**A** Maximum energy of electrons is independent of the metal type.
**B** Maximum energy of electrons is independent of the intensity of radiation.
**C** Maximum energy of emitted photocurrent is greater than that incident radiation.
**D** Maximum energy of incident radiation is less than the minimum required energy.

**35** Which graph shows how the energy, $E$, of a photon is related to its wavelength, $\lambda$?

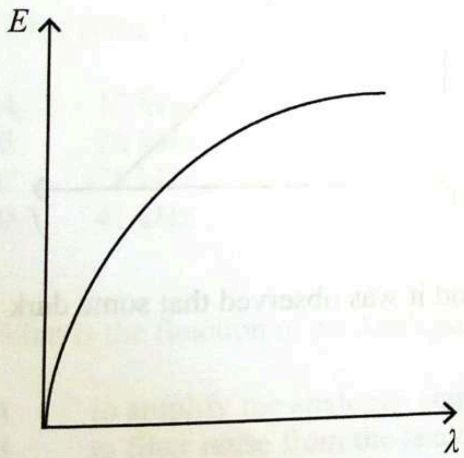
**A**



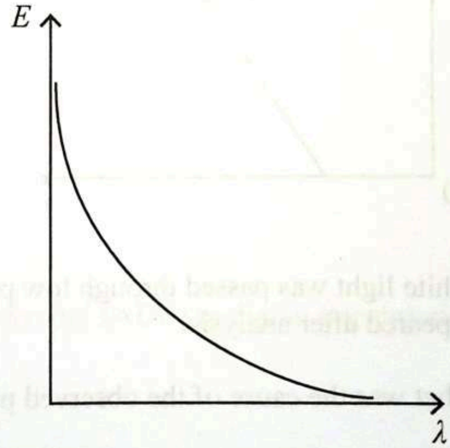

**B**




**C**




**D**

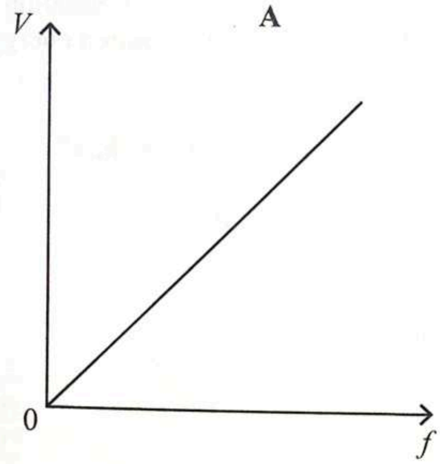




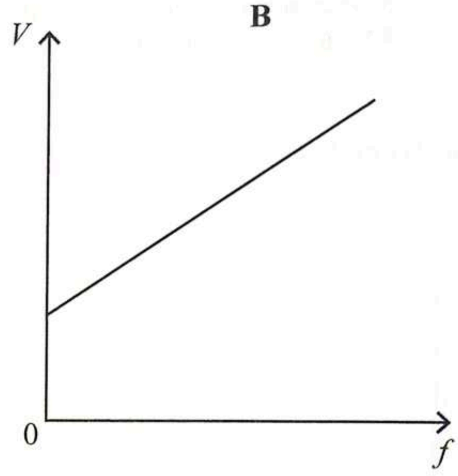

**[Turn over**

**36** Which graph shows the variation of stopping potential, $V$, with frequency, $f$, of incident radiation on a clean metal surface?

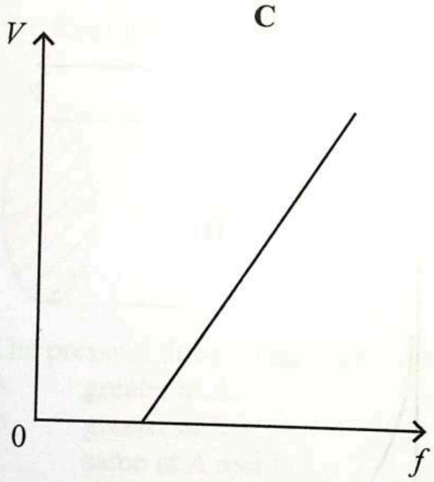
**A**



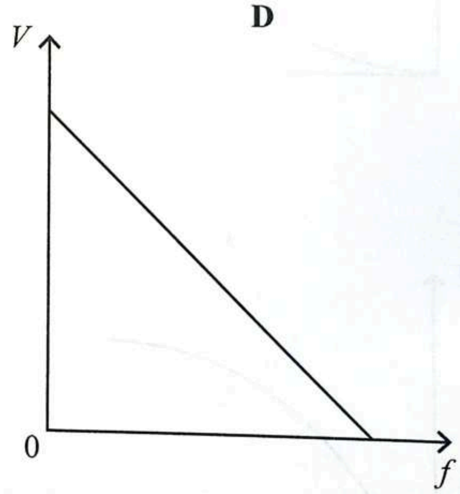

**B**




**C**




**D**




**37** White light was passed through low pressure gas and it was observed that some dark lines appeared after analysis.

What was the cause of the observed phenomenon?

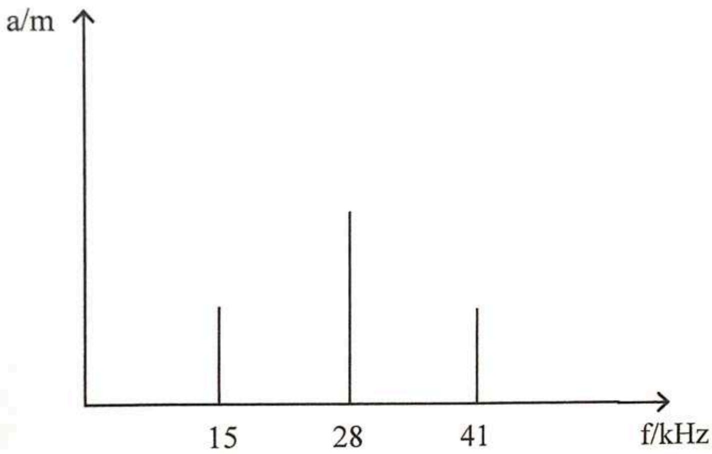
**A** some electrons casted their shadow
**B** some electrons were faster than light
**C** some electrons absorbed energy and got excited
**D** some electrons lost energy and fell to lower energy levels

38 A detector in a hospital measures the count rate from radioactive source. Fluctuations are observed in the count rate.

What do the fluctuations indicate about the nature of radioactive decay?

**A** spontaneous
**B** non linear
**C** random
**D** exponential

39 The graph shows the variation of amplitude, a, with frequency, f, of a transmitted signal.




What is the bandwidth of the signal?

**A** 13 kHz
**B** 26 kHz
**C** 28 kHz
**D** 41 kHz

40 What is the function of an Analogue to digital converter (ADC) in digital transmission?

**A** to amplify the analogue signal
**B** to filter noise from the analogue signal
**C** to convert digital signals back into analogue form
**D** to sample and quantise analogue signals into digital data

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